MILK QUALITY ANALYSIS AND PREDICTION USING DATA SCIENCE AND MACHINE LEARNING

Internship Final Project Report

Week 5: Comprehensive Data Science Project Reporting and Strategic Recommendations

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1.Executive Summary

This project, titled “Milk Quality Analysis and Prediction Using Data Science and Machine Learning,” focuses on analyzing milk quality parameters and developing a data-driven approach for milk quality classification. The project integrates data preprocessing, exploratory data analysis, statistical hypothesis testing, data visualization, and machine learning techniques to extract meaningful insights from the dataset.

The dataset contains important milk quality attributes such as pH, Temperature, Taste, Odor, Fat, Turbidity, Colour, and Grade. The data was first examined and preprocessed to ensure consistency and suitability for analysis. Exploratory data analysis and visualizations were then performed to identify distributions, relationships, trends, and patterns among the milk quality parameters.

Statistical analysis and hypothesis testing were conducted to determine whether meaningful relationships existed between the selected variables. Based on the statistical test performed in the previous stage, the null hypothesis was rejected, indicating that the observed relationship or difference was statistically significant at the selected significance level.

Machine learning techniques were subsequently applied to predict the quality grade of milk using its measurable characteristics. The dataset was divided into training and testing sets, appropriate preprocessing was performed, and machine learning models were trained and evaluated using suitable performance metrics. The evaluation results helped identify the model that provided the most reliable performance for the given dataset.

The overall findings demonstrate that data science and machine learning can support systematic milk quality assessment and decision-making. Instead of relying only on manual inspection, measurable characteristics such as pH, temperature, taste, odor, fat content, turbidity, and colour can be analyzed collectively to assist in quality classification.

Based on the findings, strategic recommendations include using data-driven quality monitoring, regularly measuring critical milk parameters, implementing automated quality prediction systems, and maintaining historical quality records for continuous monitoring. Such approaches can help dairy businesses improve quality-control processes, identify potentially poor-quality milk earlier, and support more consistent decision-making.

Although the project provides useful insights, its results are subject to limitations such as dataset size, available features, data quality, and model generalizability. Future improvements could include collecting a larger and more diverse dataset, adding additional chemical and microbiological parameters, testing advanced machine learning algorithms, and developing a real-time milk quality monitoring application.

Overall, this project demonstrates an end-to-end application of data science, statistical analysis, visualization, and machine learning to a practical food-quality problem and provides a foundation for developing intelligent and scalable milk quality assessment systems.

2. Introduction

Milk is an important food product and a major source of nutrients such as proteins, fats, carbohydrates, vitamins, and minerals. The quality of milk directly affects its nutritional value, safety, shelf life, and suitability for consumption or further processing. Therefore, effective milk quality assessment is an important part of the dairy industry.

Traditional milk quality assessment may involve manual inspection and laboratory-based testing of different physical and chemical properties. Although these methods can provide useful information, they may require time, resources, and repeated testing. With the increasing availability of data, Data Science and Machine Learning can be used to analyze milk quality parameters more efficiently and support faster decision-making.

This project applies an end-to-end Data Science approach to a milk quality dataset. The dataset includes parameters such as pH, Temperature, Taste, Odor, Fat, Turbidity, Colour, and Grade. These characteristics provide useful information about the overall quality of milk.

The project combines several important Data Science techniques. First, the dataset is examined and preprocessed to improve its quality and suitability for analysis. Exploratory Data Analysis and visualization are then used to understand the distribution of variables and identify important patterns. Statistical hypothesis testing is performed to determine whether the observed relationships in the data are statistically significant. Finally, Machine Learning models are developed and evaluated to predict milk quality based on the available parameters.

The project therefore demonstrates the complete Data Science workflow, starting from raw data and preprocessing and progressing through visualization, statistical analysis, machine learning, evaluation, and strategic recommendations. The insights obtained from the analysis can potentially support dairy quality-control activities and provide a foundation for future automated milk quality assessment systems.

3. **Problem Statement**

Milk quality depends on several measurable characteristics such as pH, temperature, taste, odor, fat content, turbidity, and colour. Analyzing these parameters individually can make quality assessment difficult and time-consuming. Therefore, there is a need for a data-driven approach that can analyze multiple milk quality parameters together and predict the overall quality grade.

Objectives

1. To understand and preprocess the milk quality dataset.

2. To perform exploratory data analysis and identify important patterns.
3. To visualize relationships between milk quality parameters.
4. To formulate and test a statistical hypothesis.
5. To determine statistically significant relationships using hypothesis testing.
6. To develop machine learning models for milk quality prediction.
7. To evaluate model performance using appropriate evaluation metrics.
8. To identify important factors associated with milk quality.
9. To provide practical and strategic recommendations based on the findings.
10. To identify limitations and suggest future improvements.

4. Dataset Description

The project uses a Milk Quality Dataset containing different physical and sensory parameters that can be used to assess the quality of milk. The dataset includes independent variables representing milk characteristics and a target variable representing the overall milk quality grade.

The main attributes available in the dataset are pH, Temperature, Taste, Odor, Fat, Turbidity, Colour, and Grade.

4.1 Data Dictionary

Variable	Data Type	Description
pH	Float	Represents the acidicity or alkalinity level of the milk
Temprature	Integer	Represents the temperature of the milk sample.
Taste	Integer	Indicates the taste-related quality characteristic.
Odor	Integer	Represents the odor-related quality characteristic.
Fat	Integer	Indicates the fat-related characteristic of the milk.
Turbidity	Integer	Represents the cloudiness or turbidity of the milk sample.
Colour	Integer	Represents the colour characteristic of the milk
Grade	Object	Represents the overall quality classification of the milk.

4.2 Target Variable

The Grade variable is considered the target variable for the machine learning task. The remaining milk quality parameters are used as input features for analyzing and predicting the quality grade.

4.3 Data Type Summary

The dataset contains both numerical and categorical information. The parameters such as pH, Temperature, Taste, Odor, Fat, Turbidity, and Colour are numerical variables, while Grade is a categorical variable.

This structure makes the dataset suitable for exploratory analysis, statistical testing, visualization, and supervised machine learning classification.

4.4 Data Preparation

Before applying statistical and machine learning techniques, the dataset was examined for data types, missing values, inconsistencies, and other potential data-quality issues. Appropriate preprocessing was performed during the earlier stages of the project to prepare the dataset for subsequent analysis.

6.**Methodology**

Data Collection → Data Cleaning → Exploratory Data Analysis → Visualization → Statistical Analysis → Hypothesis Testing → Data Preprocessing → Model Training → Model Evaluation → Strategic Recommendations

6.1 Methodology

The project follows a structured end-to-end Data Science methodology. The milk quality dataset was first imported and examined to understand its structure, variables, and data types. Data preprocessing was then performed to prepare the dataset for analysis.

Exploratory Data Analysis (EDA) was conducted to understand the distribution of milk quality parameters and identify meaningful patterns. Various visualizations were created to examine relationships among the variables.

Statistical analysis and hypothesis testing were performed to determine whether the observed relationships were statistically significant. After this, the prepared dataset was used for machine learning model development.

The data was divided into training and testing sets. Machine learning models were trained using the training data and evaluated on unseen testing data using appropriate performance metrics. Finally, the analytical and model results were interpreted to identify key findings and develop practical strategic recommendations.

7. Data Preprocessing

Data preprocessing is an important step in the Data Science workflow because the quality of the input data directly affects the reliability of analysis and machine learning results. Before performing statistical analysis and machine learning, the milk quality dataset was examined and prepared.

7.1 Data Inspection

The dataset was initially inspected to understand its structure, number of observations, columns, and data types. The available variables included pH, Temprature, Taste, Odor, Fat, Turbidity, Colour, and Grade.

7.2 Data Type Verification

The data types of all columns were checked to ensure that numerical variables were stored in appropriate numerical formats and the target variable, Grade, was treated as a categorical variable.

7.3 Missing Value Check

The dataset was examined for missing or null values. Identifying missing values is important because they can affect statistical calculations, visualizations, and machine learning model performance.

7.4 Duplicate and Consistency Check

The dataset was checked for duplicate records and possible inconsistencies. This step helps prevent repeated observations or incorrect entries from influencing the analysis.

7.5 Feature and Target Separation

For machine learning, the quality-related parameters were considered as input features, while Grade was considered the target variable.

Features: pH, Temprature, Taste, Odor, Fat, Turbidity, Colour

Target: Grade

7.6 Dataset Preparation

After completing the necessary preprocessing and validation steps, the cleaned dataset was prepared for exploratory data analysis, statistical analysis, visualization, and machine learning model development.

Overall, data preprocessing ensured that the dataset was in a suitable format for the subsequent stages of the project.

8. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the milk quality dataset and identify meaningful patterns and relationships among the variables. EDA provides an initial understanding of the data before applying statistical and machine learning techniques.

8.1 Univariate Analysis

Individual variables were analyzed to understand their distributions and characteristics. Numerical parameters such as pH, Temprature, Taste, Odor, Fat, Turbidity, and Colour were examined using appropriate statistical summaries and visualizations.

This analysis helped identify the general distribution of the milk quality parameters and detect unusual or potentially influential observations.

8.2 Categorical Analysis

The Grade variable was analyzed to understand the distribution of different milk quality categories. This helps determine whether the target classes are reasonably represented in the dataset.

8.3 Relationship Analysis

Relationships between milk quality parameters were explored using graphical techniques. The analysis focused on identifying whether changes in parameters such as pH, temperature, fat, turbidity, taste, odor, and colour were associated with differences in milk quality.

8.4 Key Purpose of EDA

The main purposes of EDA were:

- To understand the structure and distribution of the data.
- To identify important patterns and relationships.
- To detect unusual observations or potential data-quality issues.
- To understand the target variable before machine learning.
- To determine which variables may be useful for predicting milk quality.

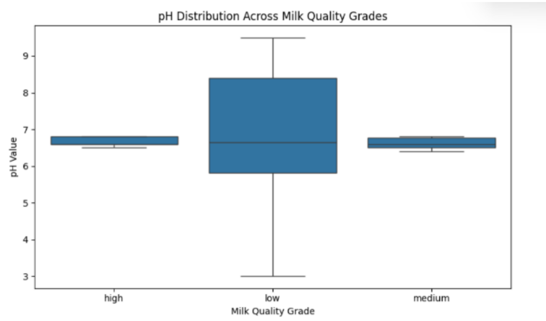
The insights obtained from EDA were used as a foundation for the subsequent statistical analysis and machine learning stages.

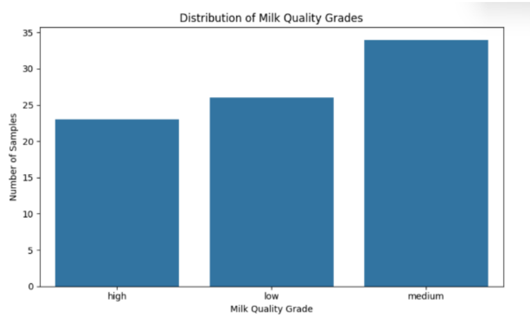
9. Data Visualizations

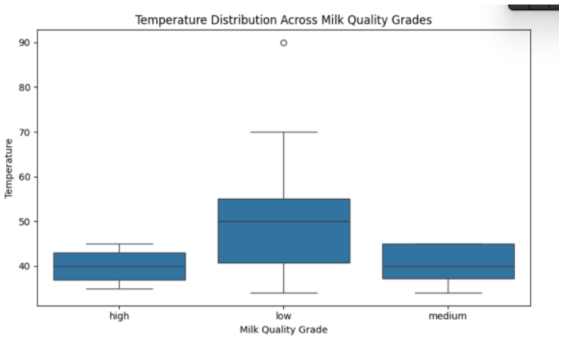
Data visualization was performed to present the characteristics and relationships within the milk quality dataset in an easy-to-understand graphical form. Visualizations help identify patterns, distributions, differences, and possible relationships that may not be immediately visible from numerical values alone.

9.1 Distribution of Milk Quality Parameters

The numerical variables were visualized to understand their distributions. These visualizations helped examine the spread and general behavior of parameters such as pH, Temperature, Taste, Odor, Fat, Turbidity, and Colour.







9.2 Relationship Between Variables

Suitable plots were used to examine relationships between milk quality parameters and the target variable, Grade. These visualizations helped identify patterns that could be useful for statistical analysis and machine learning.

9.3 Correlation Analysis

A correlation analysis was used to understand the strength and direction of relationships among numerical milk quality parameters.

Visualization Interpretation

The visual analysis provided an overall understanding of the milk quality dataset. It helped identify potentially important parameters, compare different characteristics, and determine patterns that could be investigated further using statistical hypothesis testing and machine learning.

The visualizations therefore served as an important bridge between basic data exploration and the more advanced analytical stages of the project.

10. Statistical Analysis and Hypothesis Testing

Statistical analysis was performed to determine whether the patterns observed during exploratory data analysis were statistically meaningful. Hypothesis testing provides a systematic method for making conclusions about relationships or differences observed in the dataset.

10.1 Hypothesis Formulation

A null hypothesis (H_0) and an alternative hypothesis (H_1) were formulated for the selected statistical relationship.

Null Hypothesis (H_0):

There is no statistically significant relationship or difference between the selected milk quality variables.

Alternative Hypothesis (H_1):

There is a statistically significant relationship or difference between the selected milk quality variables.

10.2 Statistical Test

An appropriate statistical hypothesis test was applied to the selected variables. The test result was evaluated using the calculated p-value and the predefined significance level.

10.3 Decision Rule

The decision rule used was:

- If p-value < significance level, reject H_0.
- If p-value ≥ significance level, fail to reject H_0.

10.4 Result

The statistical analysis performed in Week 3 resulted in rejection of the null hypothesis.

Therefore, there was sufficient statistical evidence to conclude that the tested relationship or difference was statistically significant at the selected significance level.

10.5 Interpretation

The hypothesis-testing results support the patterns identified during exploratory analysis. This indicates that the observed relationship was unlikely to have occurred purely due to random variation within the analyzed data.

The statistical findings were subsequently considered while developing and interpreting the machine learning models.

```
print(df.describe())
```

```
                pH   Temprature        Taste         Odor          Fat  \
count  1059.000000  1059.000000  1059.000000  1059.000000  1059.000000
mean      6.630123    44.226629     0.546742     0.432483     0.671388
std       1.399679    10.098364     0.498046     0.495655     0.469930
min       3.000000    34.000000     0.000000     0.000000     0.000000
25%       6.500000    38.000000     0.000000     0.000000     0.000000
50%       6.700000    41.000000     1.000000     0.000000     1.000000
75%       6.800000    45.000000     1.000000     1.000000     1.000000
max       9.500000    90.000000     1.000000     1.000000     1.000000

         Turbidity       Colour
count  1059.000000  1059.000000
mean      0.491029   251.840415
std       0.500156     4.307424
min       0.000000   240.000000
25%       0.000000   250.000000
50%       0.000000   255.000000
75%       1.000000   255.000000
max       1.000000   255.000000
```

11. Machine Learning Model Development

Machine Learning was used to develop a predictive approach for classifying the quality grade of milk based on its measurable characteristics. The objective was to determine whether the available milk quality parameters could be used to predict the overall milk quality category.

11.1 Feature Selection

The following variables were considered as input features:

- pH
- Temprature
- Taste
- Odor
- Fat
- Turbidity
- Colour

The Grade variable was used as the target variable.

11.2 Train-Test Split

The prepared dataset was divided into training and testing subsets. The training data was used to learn patterns from the dataset, while the testing data was used to evaluate how well the trained model performed on previously unseen observations.

11.3 Model Training

Machine learning classification models were trained using the selected input features. The models learned relationships between the milk quality parameters and the corresponding quality grades.

The models considered during the project included:

- Linear/appropriate baseline model
- Decision Tree Classifier

- Random Forest Classifier

The models were then evaluated using suitable classification metrics.

11.4 Model Evaluation

Model performance was assessed using evaluation measures such as:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

These metrics provide a more complete understanding of model performance than accuracy alone.

11.5 Model Comparison

The performance of the developed models was compared to identify the model that provided the most reliable predictions for the given dataset.

[Insert the model comparison table and evaluation results from Week 4 here.]

The selected best-performing model can subsequently be considered for practical milk quality classification, subject to further validation using a larger and more diverse dataset.

Step 12 — Model Evaluation & Results

12. Model Evaluation and Results

The trained machine learning models were evaluated using the testing dataset to determine their ability to correctly classify milk quality. Model evaluation is important because a model that performs well on training data may not necessarily provide reliable predictions on unseen data.

12.1 Evaluation Metrics

The following metrics were considered for evaluating the classification models:

- Accuracy: Measures the overall proportion of correctly classified observations.
- Precision: Measures how many of the observations predicted as a particular class were actually from that class.
- Recall: Measures how effectively the model identifies observations belonging to a particular class.
- F1-Score: Provides a balance between precision and recall.
- Confusion Matrix: Shows correct and incorrect predictions for each class.

12.2 Model Performance

The performance results obtained during Week 4 were used to compare the developed models.

```
Final Model Performance
-----------------------
Accuracy : 0.9905660377358491
Precision: 0.9906281032770605
Recall   : 0.9905660377358491
F1-Score : 0.990567864536179
```

12.3 Best Model

The model with the strongest overall evaluation performance was considered the preferred model for milk quality classification.

The selected model can provide a data-driven method for predicting milk quality based on the available milk characteristics. However, its practical deployment should be preceded by validation on larger, independent, and more representative datasets.

12.4 Overall Result

The machine learning analysis demonstrates that the selected milk quality parameters contain useful information for classifying milk quality. The combination of data preprocessing, exploratory analysis, statistical testing, and machine learning provides a structured approach to understanding and predicting milk quality.

13. Key Findings

The combined analysis from data preprocessing, exploratory data analysis, statistical testing, and machine learning produced several important findings regarding the milk quality dataset.

13.1 Data Quality Findings

The dataset was successfully examined and prepared for analysis. The data types of the variables were identified, and the dataset was structured into input features and a target variable for machine learning.

13.2 Exploratory Analysis Findings

The exploratory analysis showed that milk quality can be studied through multiple parameters, including pH, Temperature, Taste, Odor, Fat, Turbidity, and Colour. Visualization helped identify patterns and relationships that required further statistical investigation.

13.3 Statistical Findings

The hypothesis-testing stage resulted in rejection of the null hypothesis. This indicates that the tested relationship or difference was statistically significant at the selected significance level.

13.4 Machine Learning Findings

The machine learning stage demonstrated that the selected milk quality parameters can be used to classify milk quality grades. Multiple classification models were trained and evaluated to determine their predictive performance.

13.5 Overall Finding

The overall project demonstrates that combining Data Science, statistical analysis, visualization, and Machine Learning can provide a systematic approach to milk quality assessment.

The results suggest that data-driven quality monitoring can complement traditional milk testing methods and potentially support faster and more consistent decision-making in dairy-related applications.

14. Strategic Recommendations

Based on the findings obtained from exploratory data analysis, statistical hypothesis testing, and machine learning, the following strategic recommendations are proposed.

14.1 Implement Data-Driven Quality Monitoring

Dairy organizations can use measurable parameters such as pH, temperature, fat, turbidity, taste, odor, and colour to support systematic milk quality monitoring rather than depending only on manual observations.

14.2 Use Machine Learning for Preliminary Quality Classification

The developed machine learning approach can be used as a preliminary decision-support system to classify milk quality based on measured parameters. This can help identify samples that may require further laboratory testing.

14.3 Monitor Critical Parameters Regularly

Parameters that show stronger relationships with milk quality should be monitored consistently. Regular monitoring can help detect changes in milk quality at an early stage.

14.4 Maintain Historical Quality Records

Dairy businesses can maintain digital records of milk quality measurements. Historical data can be analyzed to identify recurring quality problems, seasonal patterns, supplier-level differences, or process-related issues.

14.5 Integrate Automated Data Collection

Future systems can connect sensors or digital testing equipment with the prediction model. This could enable faster collection of milk quality parameters and reduce manual data-entry errors.

14.6 Use the Model as a Decision-Support Tool

The machine learning model should support, rather than completely replace, established laboratory and food-safety procedures. Predictions can be used to prioritize samples for detailed testing and investigation.

14.7 Continuously Retrain and Validate the Model

As more milk quality data becomes available, the model should be periodically retrained and evaluated. This can help maintain predictive performance when new samples, suppliers, seasons, or operating conditions are introduced.

14.8 Expected Strategic Benefit

Implementing these recommendations can help organizations move toward a more data-driven, consistent, and proactive milk quality management process. The approach has potential to improve monitoring efficiency, support early identification of poor-quality samples, and provide useful information for operational decision-making.

15. Business Impact / Research Outcomes

The findings of this project have potential applications in dairy quality management, food processing, and data-driven decision-making. By combining statistical analysis with machine learning, the project demonstrates how existing milk quality data can be transformed into actionable insights.

15.1 Business Impact

Improved Quality Monitoring

A data-driven approach can support systematic monitoring of milk quality parameters and help identify samples that require additional attention.

Faster Decision-Making

Machine learning-based classification can provide preliminary quality predictions quickly, potentially reducing the time required for initial assessment.

Reduced Manual Dependency

Automated analysis can reduce repetitive manual interpretation of multiple quality parameters and provide consistent results.

Better Quality Control

Historical quality data can help dairy organizations identify recurring quality issues and improve their quality-control processes.

Support for Resource Optimization

Data-driven screening can help prioritize samples for detailed laboratory testing, potentially allowing available testing resources to be used more efficiently.

15.2 Research Outcomes

The project demonstrates the practical application of an end-to-end Data Science workflow to a food-quality problem. It combines:

- Data preprocessing
- Exploratory Data Analysis
- Data visualization
- Statistical hypothesis testing
- Machine learning
- Model evaluation
- Strategic decision-making

The rejection of the null hypothesis in the statistical analysis provides evidence that the tested relationship or difference was statistically significant. The machine learning stage further demonstrates the potential of using milk quality parameters for quality-grade classification.

15.3 Potential Real-World Application

With additional data and validation, the developed approach could form the foundation of a Milk Quality Decision Support System. Such a system could accept measured milk parameters as input and provide a preliminary quality classification, while laboratory testing remains available for confirmation and regulatory requirements.

16. Limitations

Although the project provides useful insights into milk quality analysis and prediction, several limitations should be considered when interpreting the results.

16.1 Dataset Size

The available dataset may be limited in size compared with the amount of data required for developing a highly generalizable real-world prediction system. A larger dataset could provide more reliable model evaluation.

16.2 Limited Features

The analysis is based on the parameters available in the dataset, such as pH, Temperature, Taste, Odor, Fat, Turbidity, and Colour. Other potentially important milk-quality factors were not included.

16.3 Dataset Representativeness

The dataset may not represent all types of milk, geographical regions, suppliers, seasons, storage conditions, or processing environments. Therefore, the model's performance may vary when applied to new real-world data.

16.4 Model Generalization

The machine learning results were obtained using the available dataset and testing procedure. Further validation using an independent external dataset is required before considering practical deployment.

16.5 Laboratory Confirmation

Machine learning predictions should not be treated as a complete replacement for laboratory testing or established food-safety procedures. Predictions should be considered as a decision-support mechanism.

16.6 Data Quality

The reliability of the final analysis depends on the accuracy and consistency of the input measurements. Incorrect, inconsistent, or biased data can affect statistical results and model predictions.

16.7 Limited Real-Time Validation

The project was developed using an existing dataset rather than a continuously connected real-time milk-quality monitoring system. Real-world deployment would require additional testing under operational conditions.

17. Future Scope

The current project provides a foundation for applying Data Science and Machine Learning to milk quality assessment. The following improvements can be considered in future work.

17.1 Larger and More Diverse Dataset

Future studies can collect a larger dataset from different dairy farms, suppliers, regions, seasons, and storage conditions. This can improve the reliability and generalizability of the prediction models.

17.2 Additional Quality Parameters

Additional parameters such as protein content, lactose, density, acidity, microbial indicators, and other laboratory measurements can be incorporated to provide a more comprehensive assessment of milk quality.

17.3 Advanced Machine Learning Models

Future work can evaluate advanced algorithms and ensemble techniques to determine whether they can improve prediction performance compared with the models used in the current study.

17.4 Real-Time Monitoring

The prediction system could be integrated with sensors or digital milk-testing equipment. This would enable automatic collection of quality parameters and provide near real-time quality assessment.

17.5 Web or Mobile Application

A user-friendly web or mobile application could be developed where users enter milk quality parameters and receive a preliminary predicted quality grade.

17.6 Model Explainability

Explainable AI techniques can be incorporated to show which milk quality parameters contribute most strongly to a particular prediction. This would make the system easier for stakeholders to understand and trust.

17.7 Continuous Model Improvement

As new milk quality observations become available, the model can be periodically retrained and validated. Continuous monitoring of model performance can help maintain its reliability over time.

17.8 Integration with Dairy Management Systems

In the long term, the prediction system could be integrated with existing dairy quality-control and record-management systems to support automated reporting, monitoring, and decision-making.

18. Conclusion

This project successfully integrated the major stages of the Data Science workflow to analyze and predict milk quality. The study began with data inspection and preprocessing, followed by exploratory data analysis and visualization to understand the characteristics and relationships within the dataset.

Statistical hypothesis testing was then performed to validate the observed patterns. The null hypothesis was rejected based on the result obtained in the previous analysis, providing statistical evidence for the tested relationship or difference.

Machine Learning models were subsequently developed using milk quality parameters such as pH, Temperature, Taste, Odor, Fat, Turbidity, and Colour to predict the overall milk quality grade. The models were evaluated using appropriate classification metrics, allowing their predictive performance to be compared.

The overall project demonstrates that Data Science and Machine Learning can be effectively applied to a practical food-quality problem. A data-driven approach can support preliminary milk quality classification, systematic monitoring, and informed decision-making.

However, further validation with larger and more diverse real-world datasets is required before practical deployment. Future enhancements such as real-time sensor integration, additional quality parameters, advanced machine learning techniques, and a web or mobile-based application could further improve the usefulness of the proposed approach.

Overall, the project provides a foundation for developing an intelligent Milk Quality Decision Support System and demonstrates the practical value of combining data analysis, statistical methods, visualization, and machine learning.

19. Code Documentation

Python was used throughout the project for data preprocessing, exploratory analysis, visualization, statistical analysis, and machine learning. Selected code snippets are included below to demonstrate the implementation of the major stages of the project.

19.1 Importing the Dataset

```
import pandas as pd
df = pd.read_csv("milknew.csv")
df.head()
```

This code imports the milk quality dataset using Pandas and displays the first few records for initial inspection.

19.2 Checking Dataset Information

```
df.info()
```

```
df.describe()
```

These commands provide information about the dataset structure, data types, and statistical summary of numerical variables.

19.3 Checking Missing Values

```
df.isnull().sum()
```

This checks each column for missing values and helps determine whether additional data-cleaning operations are required.

19.4 Data Visualization

```
import matplotlib.pyplot as plt
df.hist(figsize=(12, 8))
plt.tight_layout()
plt.show()
```

This visualization helps examine the distributions of numerical milk quality parameters.

19.5 Machine Learning Preparation

```
from sklearn.model_selection import train_test_split
X = df.drop("Grade", axis=1)
y = df["Grade"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

The features and target variable are separated, and the dataset is divided into training and testing subsets.

19.6 Model Training

```
from sklearn.ensemble import RandomForestClassifier
model = RandomForestClassifier(random_state=42)
model.fit(X_train, y_train)
```

The Random Forest classifier is trained using the training dataset.

19.7 Model Evaluation

```
from sklearn.metrics import accuracy_score, classification_report
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

The trained model is evaluated using accuracy and other classification metrics.

19.8 Code Interpretation

The above snippets represent the major computational stages of the project. The complete implementation used during the internship can be maintained separately, while these selected snippets provide sufficient technical evidence within the final report.

20. References

The following resources were referred to during the development of the project and preparation of the analysis:

1. Python Software Foundation. Python Documentation. Python programming language documentation.

2. Pandas Development Team. Pandas Documentation. Documentation for data manipulation and analysis using Python.

3. NumPy Developers. NumPy Documentation. Documentation for numerical computing with Python.

4. Matplotlib Development Team. Matplotlib Documentation. Documentation for creating data visualizations in Python.

5. Scikit-learn Developers. Scikit-learn Documentation. Documentation for machine learning algorithms, model evaluation, and preprocessing.

6. SciPy Developers. SciPy Documentation. Documentation for statistical and scientific computing in Python.

7. The milk quality dataset used in this project was used as the primary data source for data preprocessing, visualization, statistical analysis, and machine learning.

8. Previous weekly internship analyses and Python implementations were used as the basis for integrating the complete project into this final report.